

Virtualization and cloud computing

Before we get into the technical aspects, it's important to understand what cloud computing is actually. In fact there are many terms you come across when trying to understand the concept; such as the Internet, Grid Computing, Clusters and Cloud.

- ▶ Internet is the network of all networks connected around the world and consists of the user machines (such as your very own laptop or tablet or desktop). So no need to be confused with 'cloud' here. When people talk about internet and cloud in the same line, they're talking about cloud computing setups publicly available on the internet, nothing more, nothing less.
- ▶ Cluster is a group of computers loosely connected in a way that they can be viewed as a single system. The term is mainly used in connection with the term "Distributed Computing". Remember that what we said is about the Computing Cluster. There is another term "Data Cluster" which has a different meaning altogether.
- ▶ Grid is a term given to computers (in terms of computing resources) which are used to accomplish a common goal. Do note that a Grid can be run atop a cluster or multiple clusters and can also use resources external to those clusters.
- ▶ Cloud is a set of networked or interconnected computers which are used to provide various services. A cloud can be actually a Grid or a Cluster or a hybrid setup of a different breed. Cloud as such has an ambiguous definition because everyone perceives it in a different way and that's because it can be used in as many multiple ways.

There are a few characteristics required as such which are essential to make a networked setup of computers be called a cloud. The key features of the cloud are:

- **On-Demand:** Cloud is the set of computers which provides its services on demand, which means that the cloud can be asked for more computational resources at one time or another!
- **Self-Management:** For a computer setup to be collectively called a cloud, it should be as free from manual intervention as possible. In case of a hardware or software failure, the computers themselves should be able to figure out a way to correct the mistake and retain the services they provide.
- **Broad network access:** This is not really a different feature than a cluster or Grid, depending on how you see it. It just states that the